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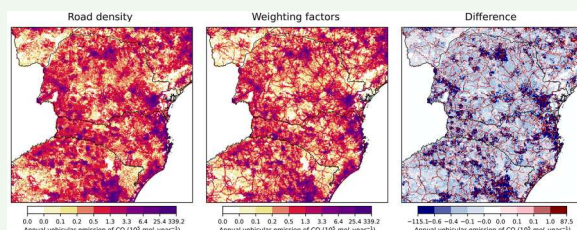
Improving spatial disaggregation of vehicular emission inventories

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ABSTRACT

Precise estimates of vehicular emissions at fine spatial scales are essential for effective emission reduction strategies. Achieving high-resolution vehicular emission inventories necessitates detailed data on traffic flow, driving patterns, and vehicle speeds for each road network segment. However, in developing countries, the lack of comprehensive traffic data, limited infrastructure, and insufficient monitoring systems constrains the development of high-resolution inventories. This gap poses significant challenges for accurately quantify emissions in regions that often experience rapid urbanisation and traffic growth. Here, we propose a novel method to enhance the spatial disaggregation of large-scale vehicular emission inventories. By analysing road-level emissions data from 63 Brazilian municipalities, we developed a model that predicts weighting factors to disaggregate vehicular emissions into a gridded format, based on the proportion of primary road lengths. Our findings indicate that the predicted weighting factors significantly improve the spatial disaggregation of vehicular emissions compared to the traditional road density method by reasonably increasing the emissions in high vehicular activity areas. This approach not only provides more accurate representations of vehicular emissions for urban planning in Brazil but also offers a solution that can be adapted to enhance top-down vehicular emissions inventories globally. Our study offers a valuable tool that can be tailored to various regions, enabling more precise urban planning and policy-making for air quality management worldwide.



Highlights

- We propose a novel method to spatially disaggregate vehicular emissions that accounts for the road type and vehicle category.
- We developed a model to generate weighting factors based on primary roads length by municipality to disaggregate vehicular emissions.
- We obtained more consistent disaggregation of vehicular emissions with the proposed method compared to road density disaggregation methodology.
- The proposed method improves PM₁₀ concentration estimation accuracy, making it valuable for air quality planning in urban areas.
- The method is well-suited for regions with limited data availability.

1. Introduction

Vehicular emissions are a leading source of urban air pollution, contributing significantly to greenhouse gas emissions and public health risks, i.e. respiratory and cardiovascular diseases [3,4,12,25,36]. Effective

control of this pollution requires refined policies and precise estimates of vehicular emissions at high spatial resolution to identify hotspots and guide environmental planning and air quality models [8,37].

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Emissions inventories play a crucial role in developing effective strategies to reduce air pollution [13], as they quantify pollutant emissions and map the spatial distribution of emission sources. However, accurately capturing the spatiotemporal variability of vehicular emissions remains challenging. These emissions are influenced by numerous factors, including vehicle type, fuel type, environmental conditions, road infrastructure, and driving behaviour, all of which can vary significantly over time and across locations [51,56].

Bottom-up methods, such as those used in MOBILE [45], MOVES [48], COPERT [30], and VEIN [23], offer high-resolution spatiotemporal data on vehicular emissions at the road level. These approaches rely on detailed traffic models and extensive input data, such as traffic flow, driving patterns, and vehicle speeds for each road segment, making them computationally intensive. While these methods provide precise results, the required data is often unavailable in many municipalities in developing countries [6,23,28].

Top-down methods, on the other hand, estimate total vehicular emissions over a larger scale, such as a municipality or national scale, using fewer input parameters compared to bottom-up methods (e.g. total fuel consumption, number of active vehicles and distance travelled by vehicle category) [38,49]. While top-down methods are less data-intensive and easier to implement, they often cannot provide detailed spatial and temporal emissions data required by air quality modellers and environmental planners [44,50].

Spatial disaggregation methods are commonly used to transform large-scale vehicular emission data into finer spatial resolutions, with normalised weighting factors (disaggregation factors) allocating emissions across grid cells [43]. The road density approach, which calculates the length of road networks within a cell relative to the total road network length across the domain, is the most widely used method. It has been applied in emissions inventories, such as the BRAZILIAN Vehicular Emissions Inventory Software (BRAVES), to spatially distribute total municipal emissions on a national scale [20]. However, the road density method tends to overestimate emissions on local roads and underestimate emissions on highways, and it is not appropriate for complex urban areas composed of interconnected centres [33,44]. Different areas, such as urban centres and industrial zones, experience varying traffic densities and, consequently, different levels of pollution. Accurate data at both small (local) and large (regional) scales allow authorities to identify pollution hotspots and implement targeted policies, such as traffic restrictions ([5]) or green infrastructures ([1]), to mitigate air pollution effectively.

Although alternative approaches incorporating traffic count data have been shown to improve disaggregation accuracy [17,39], such data is often unavailable in many municipalities, particularly in developing countries. As a result, existing disaggregation methods are often either overly data-intensive or insufficient for accurately capturing emission patterns at the road level.

To address these limitations, we developed a novel approach to improve spatial disaggregation of vehicular emission inventories that can be easily implemented using widely available data, making it more accessible for regions with limited resources. Our approach generates weighting factors based on road-level emissions inventories from three Brazilian metropolitan areas, categorised by vehicle category (i.e. passenger cars, buses, trucks, and heavy vehicles) and road type (i.e. primary and secondary roads). The novelty of our method is the use of a Generalized Linear Model to predict weighting factors for each vehicle category and municipality, based solely on primary road length proportions. We compare the proposed disaggregation method to the traditional road density approach at a 1 km x 1 km resolution and assess its impact on air quality model outcomes.

Brazil serves as an ideal case study for testing this novel approach due to its vast geographical size, diverse vehicle fleet, and varied road network characteristics. Additionally, with road transportation playing a crucial role in cargo movement and a rising vehicle ownership rate [32,40], Brazil presents a unique opportunity to examine the effectiveness of this method in real-world scenarios. The findings from this study can inform both local and global strategies for reducing vehicular emissions.

2. Methods and data

2.1. Reference emissions inventories

In this study, we used road-level vehicular emissions to derive the weighting factors. We estimated the vehicular emissions from the metropolitan area of São Paulo and Curitiba using the Vehicular Emissions Inventory (VEIN v1.0.0) developed by Ibarra-Espinosa et al. [23]. VEIN is an open-source R package that provides street-level vehicular emissions inventories with high spatial and temporal resolution based on traffic activity, vehicle fleet and emission factors input data. VEIN calibrates the activity data with the estimation of fuel consumption that matches the fuel sales in the area of study. Also, the emission factors come from the Environmental Agency of Sao Paulo ([9]) corrected with real-world emissions measurements [31].

We also included vehicular emissions data based on the inventory from Maes et al. [28]. This method was applied in the Florianópolis Metropolitan Area (FLMA) in Santa Catarina, Brazil, using a probabilistic approach that integrates fleet characteristics into emission factor estimations and employs modelled transportation data to generate a high-resolution vehicular emissions inventory. It also assessed the impact of various vehicle categories and their emissions distribution across the region.

Both inventories follow a bottom-up approach and present detailed road-level emissions by vehicle category, such as passenger cars (PC), light-duty vehicles (LDV), motorcycles (MC), buses (BUS), and trucks (TRUCKS). These sophisticated and reliable inventories served as the reference emissions for our analysis.

We considered four main pollutants to determine the weighting factors, including carbon monoxide (CO), carbon dioxide (CO₂), nitrogen oxides (NO_x), and particulate matter (PM). We classified the roads into primary (i.e. highways and high-flow roads) and secondary (i.e. residential and low-flow roads). In this work, we classified as primary roads all roads in OpenStreetMap with a speed limit above 80 km/h, as well as those tagged as 'primary,' 'secondary,' or 'trunk' roads. The remaining OpenStreetMap roads were automatically classified as secondary roads. To classify the roads from the road-level inventories, we cross-referenced the road network from those inventories with the nearest road from OpenStreetMap [16].

Figure 1 illustrates the location of the three metropolitan areas of each vehicular inventory. In total, our study assessed 63 cities, 5 of which are from the Florianópolis metropolitan area (FLMA), 19 from the Curitiba metropolitan area (CMA), and 39 from the São Paulo metropolitan area (SPMA). Table SM1 (in the supplementary material) shows vehicular emissions of CO from all metropolitan areas by road type and vehicle category. Among all regions, passenger cars had the highest emissions, while buses had the lowest emissions. FLMA and CMA showed the same rank emissions for all vehicle categories. On the other hand, SPMA had a different rank only for the LDV category, which had a higher emission compared to the MC category. Furthermore, the primary roads exhibited the highest emissions for all vehicle categories and metropolitan regions, except for TRUCKS in CMA where the emissions were higher in secondary roads.

2.2. Weighting factors model

The proposed disaggregation method uses weighting factors by vehicle category, road type, and municipality.

The weighting factors are the parcel of municipality aggregated emissions which is emitted in a road type (primary or secondary road) for each vehicle category. We estimate the weighting factors for the 63 cities using Equation 1. Where $wf_{m,r,v,p}$ is the weighting factor for the municipality m , road type r , vehicle category v , pollutant p ; $E_{m,r,v,p}$ is the emission of the municipality m , road type r , vehicle category v , pollutant p ; and $E_{m,v,p}$ is the total emission of the municipality m , vehicle category v , pollutant p .

$$wf_{m,r,v,p} = \frac{E_{m,r,v,p}}{E_{m,v,p}} \quad (1)$$

We developed a model for each vehicle category that allows to predict weighting factors for any municipality. We applied a Gaussian Generalized Linear Model (GLM) using the weighting factors from Equation 1 (dependent variable) and the proportion of primary road length (RP) from the 63 municipalities (independent variable). The road length data has been derived from the corresponding emissions inventories. Table SM3 (in the supplementary material) presents the descriptive statistics of the original dependent and independent variables of the 63 municipalities used in the model. Figure SM1 (in the supplementary material) shows the correlation matrix between independent and dependent variables considered in the models.

The linear regression equation in Equation 2 represents the wf model used in this work.

$$wf = \beta_0 + \beta_1 * RP \quad (2)$$

where wf is the weighting factor by vehicle category (the dependent variable), β_0 represents the predicted value of the dependent variable when the independent variable is equal to zero (intercept); β_1 is the model coefficient of the independent variable; and RP is the proportion of primary road length (independent variable). In this study, the residuals of the models for each vehicle category satisfied the assumptions of independence, homoscedasticity, and normality ([53]), as shown in Figure SM2 (in the supplementary material).

To analyse the performance of the models to predict the weighting factors, we applied a leave-one-out cross-validation [27]. Hence, we conducted 63 iterations of the GLM, each iteration removing one municipality from the sample. For each iteration, we applied the model to calculate the weighting factor for the previously removed municipality. We then calculated the percentage error between the model's weighting factor value (Equation 2) and the observed weighting factor value (Equation

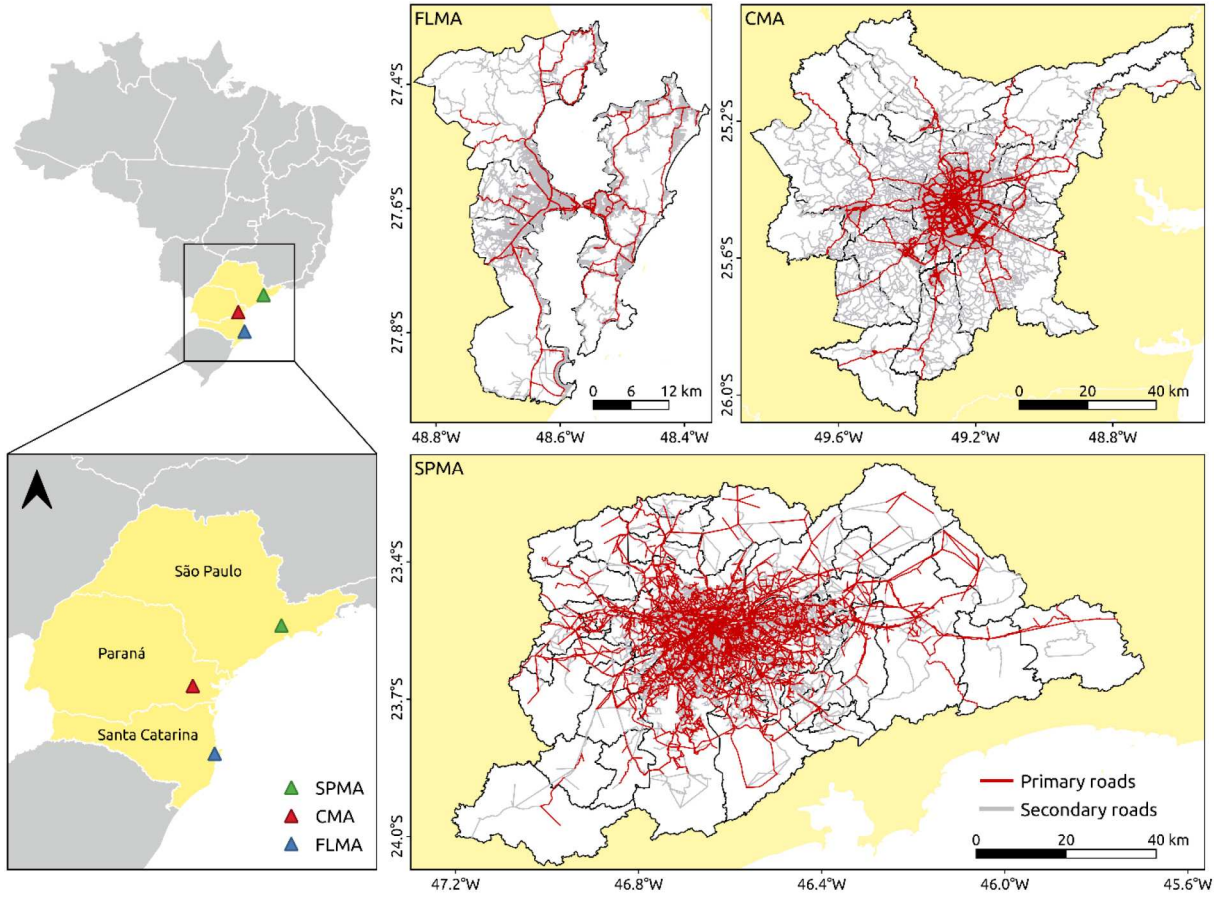


Figure 1. Metropolitan areas assessed in this study. SPMA is the São Paulo metropolitan area, CMA is the Curitiba metropolitan area and FLMA is the Florianópolis metropolitan area.

1), as shown in Equation 3.

$$\text{Error \%} = \left| \frac{wf_{\text{observed}} - wf_{\text{predicted}}}{wf_{\text{observed}}} \right| * 100 \quad (3)$$

2.3. Evaluation of weighting factors methodology

The spatial resolution of 1km² has been used in this work to disaggregate emissions from the metropolitan areas. This resolution is considered high enough to find air pollution hotspots [35], used as input in air quality models [11,26,33] and detect mesoscale circulations [15]. To disaggregate the emissions from each municipality of the metropolitan areas we used Equation 4, where $E_{m,p,v,r,c}$ is the emission in the municipality m of pollutant p , vehicle category v , road type r in the grid cell c ; $E_{m,p,v,r}$ is the total emission in the municipality m , pollutant p , vehicle category v and road type r (aggregated emissions by municipality from references inventories); $L_{c,m,r}$ is the length of roads in the grid cell c at the municipality m by road type r (primary or secondary roads); $L_{m,r}$ is the total length in the municipality m of the road type r ; and $wf_{v,r}$ is the predicted weighting factor by vehicle

category v and road type r obtained through the models (Equation 2).

$$E_{m,p,v,r,c} = E_{m,p,v,r} * \frac{L_{c,m,r}}{L_{m,r}} * wf_{v,r} \quad (4)$$

To evaluate the accuracy of the proposed disaggregation method, we aggregated the reference emissions by road type into cells with 1 km x 1 km of resolution. These aggregated emissions (called reference emissions) were compared with disaggregated emissions using the weighting factors (the proposed methodology of this work) and using the standard road density method reported by Tuia et al. [44]. These comparisons were made to evaluate the accuracy of the proposed method against the road density method.

We computed the Fractional bias, bias, Spearman correlation, Hellinger distance, and acceptability criteria to evaluate the model's performance. The Fractional bias was calculated according to Equation 5 [10], where $FB_{p,c}$ is a fractional bias of the pollutant p in cell c ; $E_{d,c}$ is the emission from the disaggregated method d (weighting factors or road density) into the cell c and $E_{ref,p,c}$ is the reference emission of pollutant p into the

cell c . Negative values of bias and fractional bias indicate an underestimation of disaggregated emissions in relation to reference emission, while positive fractional bias indicates an overestimation. We used the Spearman index to measure the correlation between our method and the reference. Acceptability criteria were calculated to verify which cells fulfilled a certain threshold [54]. A threshold of 50% was applied for the absolute difference between disaggregation emissions and reference emissions in each cell. A statistical similarity, namely the Hellinger distance (D_h), was applied to compare the reference emission with both disaggregation methods as shown in Equation 6 [29], where $E_{d,c}$ is the disaggregated emission using disaggregation method d into the cell c and $E_{ref,c}$ is the reference emission into the cell c . The Hellinger distance serves as a similarity measure, reflecting the correlation between two statistical distributions, in which a value closer to zero indicates a higher similarity between the distributions. All statistics used in this work have been used in similar studies to evaluate disaggregation methods [17,33,39,44]. Analysis was conducted comparing the performance of both disaggregation methods with the reference emissions.

$$FB_{p,c} = \frac{(E_{d,p,c} - E_{ref,p,c})}{0.5*(E_{d,p,c} + E_{ref,p,c})} \quad (5)$$

$$D_h = \left(\sum_{c=1}^n \left(\sqrt{E_{d,c}} - \sqrt{E_{ref,c}} \right)^2 \right)^{\frac{1}{2}} \quad (6)$$

2.4. Implementing weighting factors into BRAVES

The BRAZilian Vehicular Emissions inventory Software (BRAVES) database estimates vehicular emissions using a probabilistic method, in which users can define the desired spatial and temporal resolution. The BRAVES database provides hourly and annual emissions of 41 gases and particles. The software was developed in Python and provides a vehicular emissions database for Brazil using simple input data at the municipality level [20,49]. BRAVES delivers vehicular emissions data with spatial disaggregation based on the original road density, temporal disaggregation using vehicle flow profiles.

In this work, we implemented our proposed method into BRAVES in order to compare it with the original estimates from BRAVES based on road density. We estimate the weighting factors, using the new approach for the municipalities within a domain located in Southern Brazil, as shown in Figure SM8 (in the supplementary material). For this larger domain, we considered a spatial resolution of 4 km x 4 km. In this case, we utilised

the road network data from OpenStreetMap to establish RP by municipalities as independent variables in our models. In this case, where reference emission is unavailable, the comparison between both disaggregation methodologies allowed us to assess the spatial variations in the emissions.

2.5. Coupling vehicular emission inventories into CMAQ

We used the Community Multiscale Air Quality (CMAQ) model, version 5.3.2 [46] to simulate atmospheric concentrations of CO, PM, nitrogen dioxide (NO₂) and ozone (O₃). CMAQ is an advanced atmospheric modelling system that integrates air quality and meteorological data to simulate the transport, transformation, and removal of pollutants across multiple spatial scales. It also allows for the comprehensive simulation of both gas-phase and particulate pollution processes, making it a valuable tool for understanding air quality impacts. In this study, we employed the Carbon Bond 06 (CB6) [55] and AE7 mechanisms [14] to simulate gas-phase chemical mechanisms, aerosol formation, and dynamic processes. The modelling domain is shown in Figure SM9 in the supplementary material. Domain d01 covers Brazil with a spatial resolution of 20 km x 20 km. This larger domain provides boundary conditions for the smaller domain d02, which has a spatial resolution of 4 km x 4 km. We considered 21 days of hourly simulation, starting on December 31, 2018, at 10:00 PM and ending on January 21, 2019, at 10:00 PM. The initial two hours of the simulations were dedicated solely to stabilisation and were subsequently excluded from the analysis.

The input data for each domain is derived from a diverse range of models and sources. CMAQ inputs are broadly categorised into meteorological conditions, chemical boundaries, and emissions. Meteorological conditions for both domains were simulated using Weather Research and Forecasting Model (WRF) version 4.0. We conducted WRF simulations in 36-hour cycles, allocating the initial 12 hours for model stabilisation, and subsequently discarding them before using the remaining 24 hours as input into CMAQ. Each simulation extends for 24 hours with a time step interval of 1 hour, with the final hour of the preceding simulation serving as the initial condition for the subsequent one.

To achieve an accurate representation of atmospheric composition, we incorporated multiple emission inventories: biomass burning emissions from the Fire INventory from NCAR (FINN) version 1.5 [52], industrial emissions from Instituto de Meio Ambiente [24], biogenic emissions from the Model of Emissions of Gases

and Aerosols from Nature (MEGAN) version 3.2 [19,41], and vehicular emissions from BRAVES [49]. Further details are available in Hoinaski et al. [21].

We created two simulation scenarios by incorporating different vehicular emission inventories as input data for CMAQ: one based on the weighting factors disaggregation method (WF) proposed in this study, and the other based on the road density disaggregation method (RD). Emission inventories from all sources were integrated into domain d01 with a spatial resolution of 20 km x 20 km and domain d02 with a spatial resolution of 4 km x 4 km.

We compared both CMAQ-modelled scenarios (WF and RD) with data from 12 air quality monitoring stations located in the states of Rio Grande do Sul, Santa Catarina, and Paraná, within domain d02 as shown in Figure SM9. The aim of this comparison is to assess the effectiveness of integrating each vehicular emission inventory into CMAQ for simulating air pollution concentrations that closely match the observed concentrations from the monitoring stations. To achieve this, we selected cells from the CMAQ simulations within a 0.1-degree radius of each monitoring station to calculate bias, Spearman correlation, root mean squared error (RMSE), and mean absolute error (MAE) using the observational data from the monitoring stations.

3. Results and discussion

3.1. The vehicular emissions weighting factor model

Table SM2 in the supplementary material presents the weighting factors of CO, CO₂, PM and NO_x for primary roads by municipality and vehicle category. Across these pollutants, no statistically significant differences in weighting factors were observed, indicating that the emission proportions for primary and secondary roads remain consistent regardless of pollutant type. Furthermore, light vehicle categories (i.e. PC, LCV, and MC) showed the same weighting factors, suggesting similar emission patterns for light vehicles on both primary and secondary roads. In a previous study, Ossés de Eicker et al. [33] identified minor variations in the spatial distribution when changing pollutants. Therefore, our analysis focuses on the weighting factors for CO, considering the following vehicle categories: PC,

BUS, TRUCKS and HEAVY, where HEAVY category is the combined emissions of BUS and TRUCKS.

Table 1 shows the parameters estimated by the GLM for each vehicle category with their respective confidence interval and *p*-value. Confidence intervals provide a range of values within which the true population estimate is likely to fall, with a specified level of certainty (95% in this study). This means that if the study were repeated, we would expect the true value to fall within this range 95% of the time. The *p*-value, on the other hand, indicates the probability that the observed results occurred by chance. In this analysis, a *p*-value below 0.05 is considered statistically significant, suggesting less than a 5% likelihood that the results are due to random variation alone. All vehicle categories have shown statistically significant model coefficients (β_0 and β_1). Furthermore, Figure SM3 in the supplementary material demonstrates the weighing factors (wf) observed against predicted with the coefficient of determination (R^2) of the models by vehicle category. The R^2 provides a measure of how well observed outcomes are replicated by the model, based on the proportion of total variation of outcomes explained by the model. The HEAVY category shows the highest R^2 value at 0.36, followed by TRUCKS (0.22), BUS (0.19), and passenger cars (0.09). These results suggest that the model's predictive accuracy varies considerably across vehicle categories, with better performance for heavier vehicles.

Table 2 presents the percentage error of the models using leave-one-out cross-validation by vehicle category, as shown in Equation 3. The leave-one-out cross-validation analysis indicated a notable mean error across all categories. Nonetheless, the median error remained relatively low, with PC category at 20% and HEAVY category at 24%. TRUCKS and BUS categories exhibited median errors of 26% and 27%, respectively. It suggests that although the model presented large errors for a few municipalities, it generally had smaller errors for most municipalities.

3.2. Comparison between reference emissions and disaggregation methods

Our results reveal that using weighting factors to disaggregate vehicular emissions is in closer agreement with the reference emissions than the road density method.

Table 1. Generalised linear model parameters with their respective confidence interval (CI) of 95% and *p*-value.

Categories	β_0	CI (95%)	<i>p</i> -value	β_1	CI (95%)	<i>p</i> -value
PC	0.418	(0.307; 0.528)	1.61E-13	0.364	(0.096; 0.632)	7.84E-03
BUS	0.344	(0.221; 0.466)	3.64E-08	0.600	(0.305; 0.896)	6.89E-05
TRUCKS	0.318	(0.190; 0.446)	1.13E-06	0.687	(0.377; 0.996)	1.34E-05
HEAVY	0.259	(0.156; 0.363)	8.99E-07	0.767	(0.517; 1.017)	1.86E-09

β_0 is the model intercept parameter; β_1 is the model coefficient of the independent variable (RP).

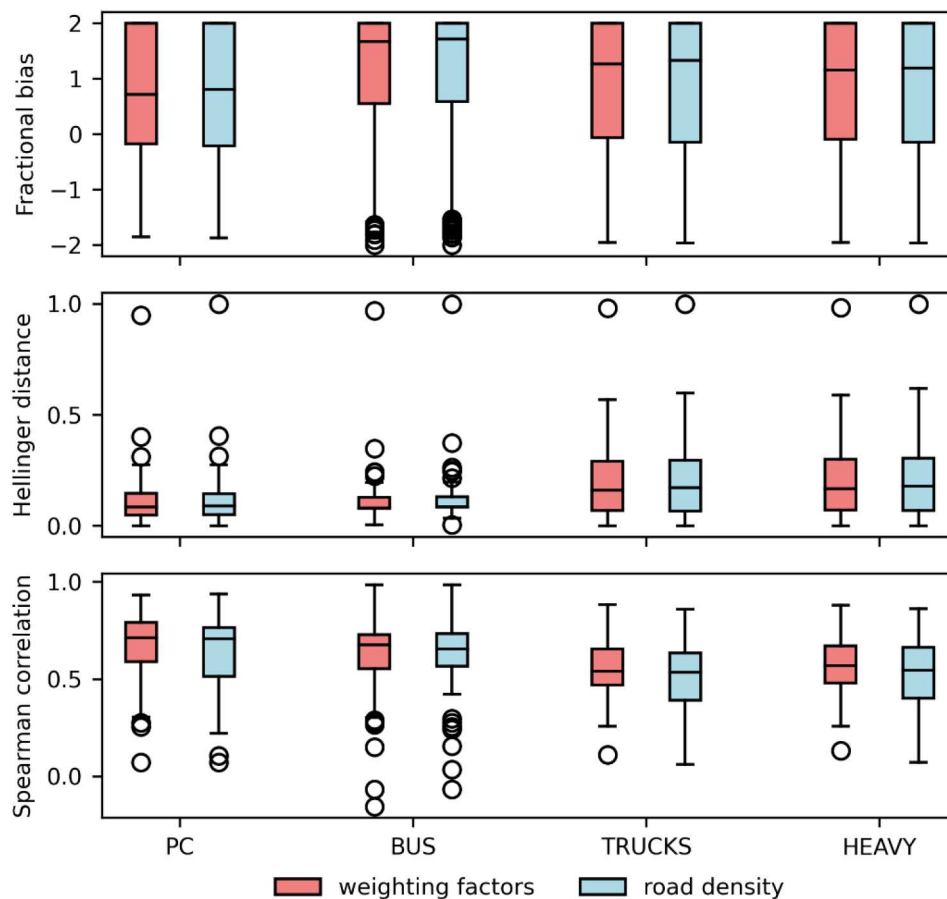
Table 2. Percentage error (%) of the models using leave-one-out cross-validation (Equation 3).

Category	Mean	Median	Minimum	Maximum	Interquartile range
PC	1011	20	2	59032	29
BUS	94	27	1	2190	36
TRUCKS	113	26	1	1605	52
HEAVY	65	24	1	706	39

Figure 2 shows the variation of fractional bias, Hellinger distance, and Spearman correlation between the disaggregation methods against the reference. Although both disaggregation methods have shown similar results for fractional bias, the weighting factors method slightly decreased the median fractional bias in comparison with the road density method, for all vehicle categories. Similarly, the Hellinger distance is also reduced with the weighting factors method. In addition, weighting factors increased the median Spearman correlation for all vehicle categories and reduced the interquartile range for PC, TRUCKS, and HEAVY categories.

To analyse the influence of road type in both methods, we compared the reference emission from primary and secondary roads with the disaggregated emissions (i.e. weighting factors and road density

disaggregation methods) by road. Figure 3 shows the fractional bias boxplot and the scatterplot of CO emissions by road type for both disaggregation methods for passenger cars and heavy vehicle categories. The weighting factors method has presented a lower median fractional bias in secondary roads when compared with the road density method. However, for primary roads, the median fractional bias is higher, and the interquartile range is lower for the weighting factors method. In addition, road density underestimates the emissions in primary roads while weighting factors slightly overestimate. The fractional bias has a similar pattern for all vehicle categories (Figure 3 and Figure SM4 in the supplementary material). Emissions disaggregated by weighting factors showed a tendency to be slightly closer to the reference line 1:1. Moreover, the weighting factors method presented the highest percentage of cells within the acceptance criteria (F2) for all categories, of which PC and HEAVY showed the highest percentage for primary roads, 39% and 30%, respectively (Figure 3). For BUS, the weighting factors increased by 4% of the percentage of cells within the acceptance criteria in secondary roads while increasing

**Figure 2.** Fractional bias, Hellinger distance, and Spearman correlation between the reference emission and both disaggregation methods.

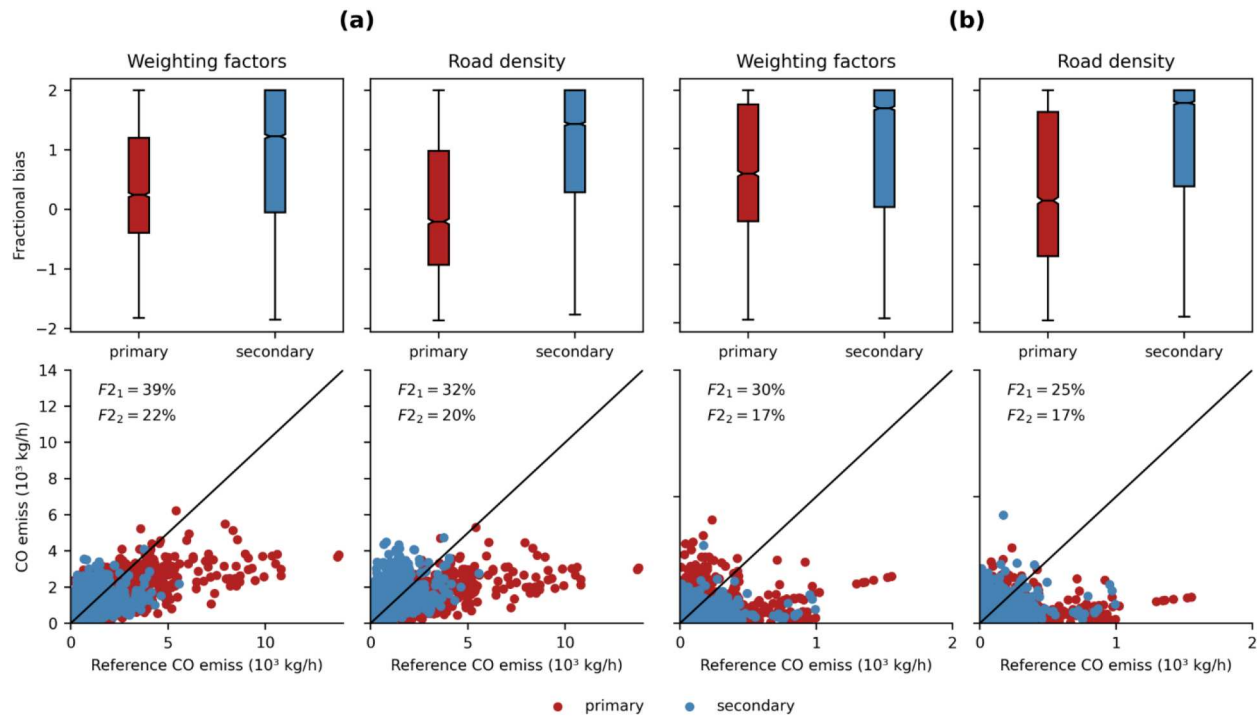


Figure 3. Fractional bias boxplot and CO emissions scatterplot by road type for (a) PC and (b) HEAVY for each disaggregation method. The diagonal 1:1 line in the subplot represents the same emissions for reference and disaggregated methods.

by 4% for TRUCKS in primary roads (Figure SM4 in the supplementary material).

Table 3 illustrates the percentage of municipalities in which the weighting factors disaggregation method improved statistical metrics by vehicle category and road type. We observed an increase in the percentage of municipalities which has improved the statistical metric after using the weighting factors. Considering all roads (total road), the weighting factors improved for more than 50% of the cities for all vehicle categories.

Table 3. Percentage of municipalities (%) which the weighting factors disaggregation method improved bias, fractional bias, Spearman correlation, acceptance criteria, and Hellinger distance when compared with the reference emissions by vehicle category and road type.

Statistics	Road type	PC	BUS	TRUCKS	HEAVY
Bias*	Total	81	61	73	68
Fractional bias	Total	59	59	51	49
Acceptance criteria	Total	54	46	60	59
Hellinger distance	Total	70	57	75	73
Correlation	Total	71	56	76	75
Bias*	Primary	49	16	33	35
Fractional bias	Primary	43	11	24	27
Acceptance criteria	Primary	66	69	63	64
Hellinger distance	Primary	49	24	33	33
Bias*	Secondary	81	90	95	95
Fractional bias	Secondary	84	100	98	98
Acceptance criteria	Secondary	39	41	53	51
Hellinger distance	Secondary	79	90	89	92

*All statistics were analysed on average, except for bias, which was analysed using the median.

We do not observe improvement for BUS and HEAVY categories when analysing the acceptance criteria and fractional bias, respectively. For primary roads, the weighting factors method improved the acceptance criteria by up to 63% for all vehicle categories. On the other hand, the use of weighting factors in primary roads did not improve bias, fractional bias, and Hellinger distance in most municipalities. In contrast, the use of weighting factors in secondary roads has improved bias, fractional bias, and Hellinger distance in more than 79% of the municipalities for PC, 90% for BUS, 89% for TRUCKS, and 92% for HEAVY. However, the acceptance criteria improved in 53% of the municipalities for TRUCKS and 51% for HEAVY and less than 50% for PC and BUS. Table SM4 in the supplementary material shows the percentage of cells within the overall acceptance criteria considering all cells from all municipalities.

In all metropolitan areas, the road density method showed a consistent pattern of underestimating emissions in highways and overestimating emissions in residential zones, which is aligned with findings from previous studies [33,44]. The weighting factors method slightly minimised those underestimations in highways, which were more visible in FLMA, where cells from highways changed into lower fractional bias values. In addition, cells with secondary roads (residential zones) are still overestimated by using weighting factors, mainly in roads with low emissions. Figure 4 shows the

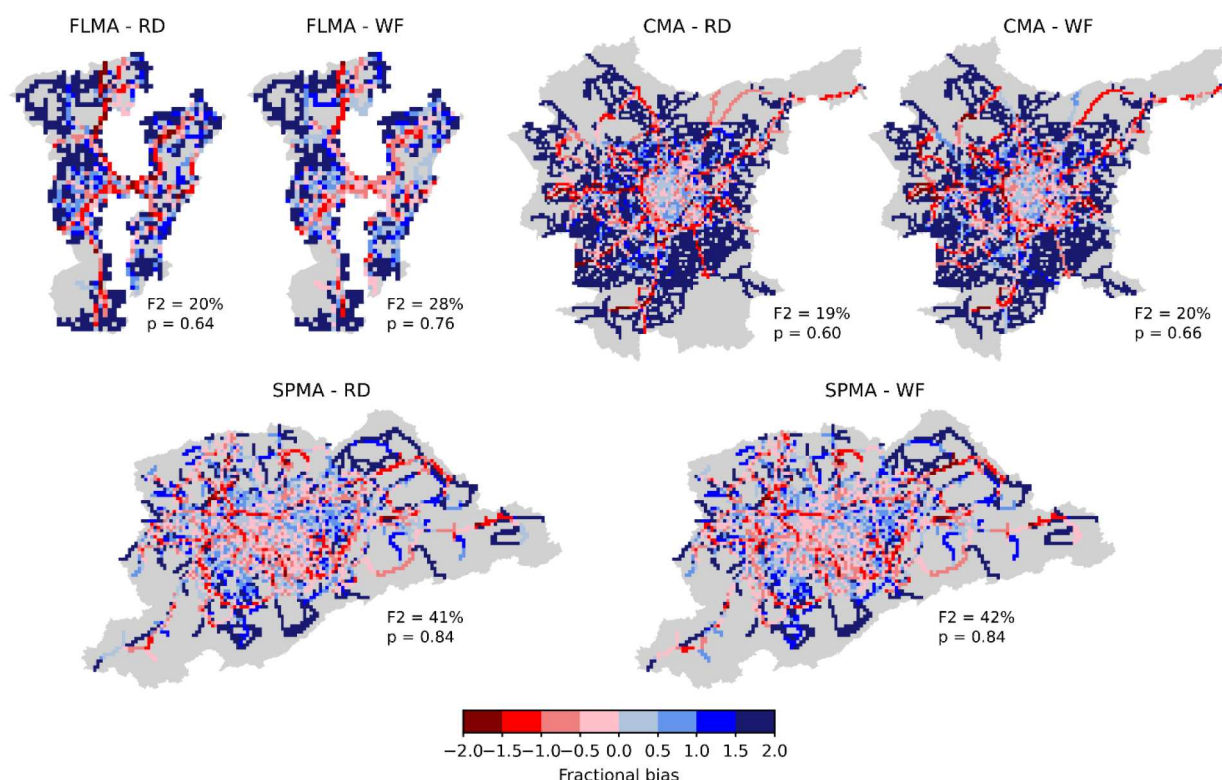


Figure 4. Fractional bias of CO emissions for passenger cars between the reference and disaggregated emissions in each metropolitan area analysed. RD is the road density method, WF is the weighting factors method, FLMA is the Florianópolis metropolitan area, CMA is the Curitiba metropolitan area, SPMA is the São Paulo metropolitan area, F2 is the percentage of cells within the acceptance criteria and p is the Spearman correlation.

fractional bias of CO emissions for passenger cars, comparing the reference emissions with the emissions estimated by both disaggregation methods across each metropolitan area analysed. For PC, the Spearman correlation and the percentage of cells within the acceptance criteria increased by using the weighting factors method in FLMA and CMA. On the other hand, SPMA showed less difference between both disaggregation methods. Figure SM5 to Figure SM7 (in the supplementary material) illustrate the fractional bias maps, percentage of acceptance criteria, and Spearman correlation for TRUCKS, HEAVY, and BUS categories. FLMA exhibited similar results for all vehicle categories, wherein the weighting factors method increased the percentage of cells within the acceptance criteria and Spearman correlation. In SPMA, no discrepancies were observed in BUS emissions when comparing both disaggregation methods. However, an increase in the Spearman correlation was observed for HEAVY and TRUCKS emissions. Furthermore, the acceptance criteria improved for TRUCKS while remaining unchanged for HEAVY. In relation to CMA, we found similar results for HEAVY and TRUCKS, in which the weighting factors increased Spearman correlation, although it led to a slight decrease in the percentage of cells meeting the

acceptance criteria. Among the 19 municipalities analysed in the CMA, BUS emissions were observed in only 5 municipalities, as shown in Figure SM7 (in the supplementary material). For BUS in CMA, the use of the weighting factors method enhanced the percentage of cells within the acceptance criteria, while the Spearman correlation remained unchanged compared to the road density method.

3.3. Implementation of weighting factors model into BRAVES

Figure 5 shows the implementation of the weighting factors models by vehicle category in the municipality-level emissions from BRAVES in Southern Brazil in a spatial resolution of 4 km x 4 km. We observed a visually enhanced and more realistic scenario for CO emissions using the weighting factors model compared to the road density method. The difference between both disaggregation methods reveals that the use of weighting factors results in a reduction of emissions within urban centres and a redistribution of those emissions to primary roads (i.e. highways) that connect urban areas. The Mann–Whitney test was applied to disaggregated emissions. The result shows that there is a significant

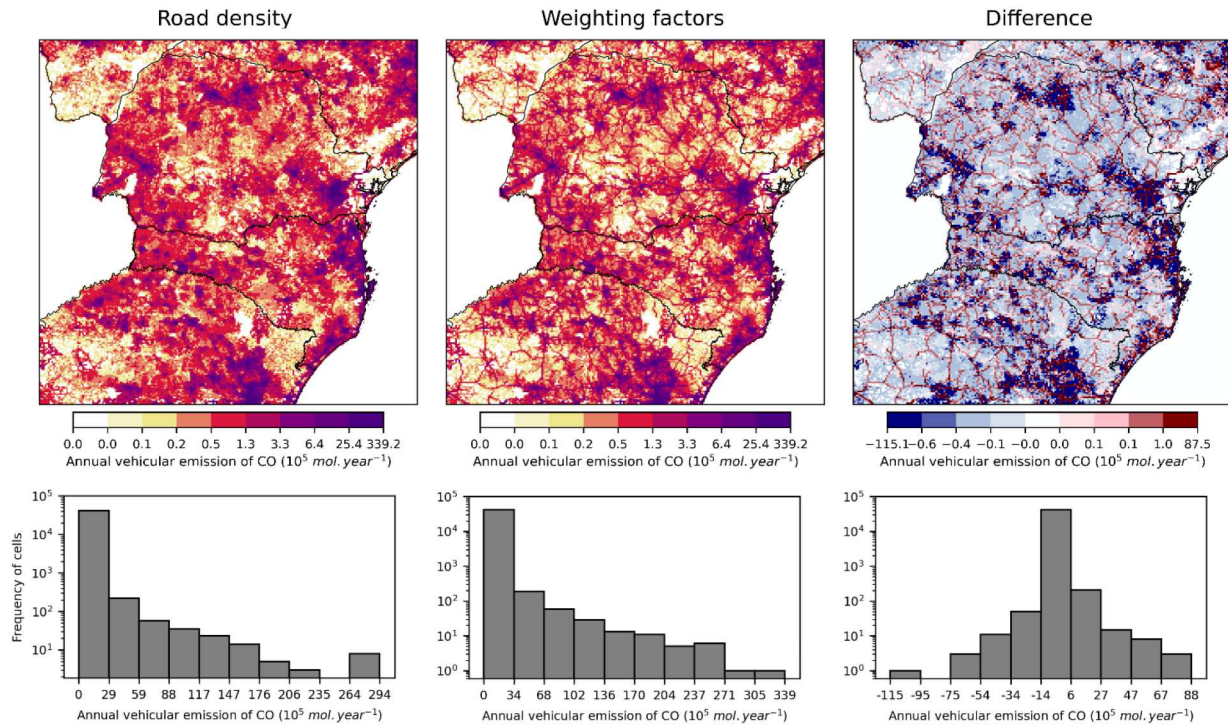


Figure 5. CO emissions from all vehicle categories at the municipality level from BRAVES [49] disaggregated by road density and weighting factors models (Equation 2 using the parameters of Table 1), and CO emission difference between both disaggregation methods in a spatial resolution of 4 km x 4 km. A positive difference indicates an overestimation of the weighting factors method.

difference between the distributions of CO emissions from both disaggregation methods (p -value = 4.75E-54).

Alam et al. [2] developed a robust method to distribute vehicular emissions at the county level into road-level scale based on traffic volume data. The study compared the proposed methodology based on traffic volume data against the road density disaggregation methodology. The authors found that road density methodology concentrates the emissions only in central areas, while high-traffic roads and low-traffic roads receive similar weight. On the other hand, the disaggregation methodology based on traffic volume resulted in higher vehicular emission estimations on highways and major roads. Despite not including traffic volume data, the disaggregation method proposed in this work yields emissions results that closely resemble those reported by Alam et al. [2], where the disaggregation emissions are higher in high-traffic roads (i.e. highways) than in low-traffic roads.

Our findings align with those from previous studies that highlight the limitations of traditional road density-based methods for emissions disaggregation, particularly in complex urban environments. For example, Gómez et al. [17] and Saide et al. [39] demonstrated that incorporating detailed traffic count classifications and road capacity data improves emissions estimates in high-traffic urban zones compared to

simpler top-down methods. Similarly, Osses de Eicker et al. [33] and Tuia et al. [44] found that while road density-based disaggregation works reasonably well in small to mid-sized cities, it often underestimates emissions in city centres and industrial zones of larger urban areas. By using weighted factors based on vehicle category and road type, our disaggregation method offers enhanced spatial resolution that better captures vehicular emissions along primary roads and high-emission zones.

In this work, we have applied samples from Brazilian municipalities to develop a statistical model for predicting disaggregation factors. We provide confidence intervals for the models' parameters (Table 1) and the percentage error of the weighting factors predictions (Table 2). Due to the inherent properties of the statistical model, our method could reproduce a higher level of uncertainty in the estimation of the weighting factors. Therefore, we recommend the usage of our method only for disaggregating emissions from municipalities with similar characteristics to our study.

Key characteristics that are crucial for the weighting factors method applicability include a complex road network, diverse vehicle fleet composition, and high traffic density, as seen in the metropolitan areas studied. These regions also have well-defined primary and secondary road classifications and significant data

availability on traffic patterns and emissions, which contribute to the model's accuracy. Municipalities with simpler road networks, lower traffic volumes, or less varied vehicle types may not yield equally reliable results, as these factors play a critical role in our model's predictive accuracy. By considering these specific characteristics, users can better assess whether this method is suitable for other regions with comparable urban infrastructure and traffic patterns.

3.4. Performance of the coupling of vehicle emission estimates into CMAQ

We assessed the performance of integrating vehicular emissions estimates into CMAQ by comparing modelled air pollutant concentrations (CMAQ model output) with observational data from air quality monitoring stations within a domain in southern Brazil. We simulated atmospheric pollutant concentrations using two distinct vehicular emission inventories as input data in CMAQ: (i) the weighting factors (WF) method and (ii) the road density (RD) method, thereby creating two different scenarios. Table SM4 in the supplementary material presents the mean values of bias, Spearman correlation, RMSE, and MAE for both scenarios simulated by CMAQ compared to the monitoring station data.

The WF scenario showed lower bias, RMSE, and MAE values for NO₂ at 7 out of 9 monitoring stations (PR03, RS02, RS03, RS05, RS07, RS09, SC03). However, both scenarios showed similar Spearman correlations (ρ). The WF scenario produced higher errors (bias, RMSE, MAE) at monitoring station SC01 compared to the RD scenario. For ozone, 4 out of 8 monitoring stations exhibited lower bias under the WF scenario (PR03, RS05, RS09 and SC03). Despite a higher bias at SC01, the WF scenario showed lower RMSE and MAE than the RD scenario. In terms of CO, the WF scenario resulted in lower bias, RMSE, and MAE at monitoring station RS03, and higher Spearman correlation compared to the RD scenario. Although the WF scenario showed higher bias at RS07, it also displayed lower RMSE and MAE, along with a higher Spearman correlation. Regarding PM₁₀, the WF scenario had lower bias and MAE at 4 out of 8 monitoring stations (RS05, RS07, SC01, and SC03). Additionally, the WF scenario showed lower RMSE at 6 out of 8 stations (RS02, RS05, RS07, SC01, SC02, and SC03). Conversely, the WF scenario exhibited high errors at station RS09, likely influenced by its location, which is farther from the roads.

Figure SM10 in the supplementary material presents the boxplot of the mean values of bias, Spearman correlation, RMSE, and MAE resulting from the comparison between the monitoring station data and the two

scenarios modelled by CMAQ: WF and RD. On average, considering the metrics of all monitoring stations combined, we observed small differences between the two modelled scenarios. However, the largest differences were observed for NO₂, where the WF scenario had a lower median correlation and higher median bias and MAE compared to the RD scenario. Additionally, the WF scenario showed a lower median Spearman correlation and higher median RMSE and MAE for CO. On the other hand, for PM₁₀, the WF scenario revealed lower median bias, RMSE, and MAE compared to the RD scenario. Both scenarios produced similar results for ozone, indicating that using different vehicular emissions inventories as input into CMAQ does not affect ozone modelling.

The varying performance of the WF method across pollutants can be attributed to the specific characteristics of each pollutant's emissions and dispersion. The improved performance for PM₁₀ may be because PM₁₀ emissions are more directly linked to vehicular activity and road dust resuspension [34], which are well-represented by the weighting factors method. In contrast, NO₂ and CO emissions are more influenced by localised traffic conditions, vehicle operation modes (e.g. acceleration and idling), and proximity to emission sources, making these pollutants more challenging to model accurately [7]. The weighting factors method may not fully capture these localised variations, leading to higher errors for NO₂ and CO. Moreover, NO₂ is subject to rapid chemical transformations in the atmosphere, which adds complexity to accurately predicting its concentrations with a general emissions model. The lower median bias, RMSE, and MAE for PM₁₀ in the WF scenario suggest that the weighting factors methodology is particularly suited for pollutants with more consistent spatial distributions related to road density and vehicle type.

4. Conclusion

This work presents a novel approach by developing a model to predict weighting factors to spatial disaggregate vehicular emissions, using Generalized Linear Models (GLMs) across four vehicle categories (i.e. passenger cars, buses, trucks, and heavy). The GLM models consider the proportion of primary road length by municipality as the independent variable to predict weighting factors, making it adaptable for use in municipalities with similar data availability. This study is also unique in its integration of disaggregation methodologies directly into an air quality model (CMAQ), allowing to assess the method's effectiveness for simulating pollutant distributions.

Our results revealed an improvement in the disaggregation of vehicular emissions using the weighting factors method compared to the traditional road density method. The weighting factors from our method slightly decrease fractional bias and Hellinger distance. We observed an increase in Spearman correlation and a decrease in the interquartile range using the weighting factors models. These improvements have been observed for most municipalities in the study area.

The implementation of weighting factors models into BRAVES resulted in a more accurate estimate of vehicular emissions. The new method effectively represented the interconnectivity between cities and significant emissions in highways, which would not be perceived when employing the road density method. The Mann–Whitney test revealed a statistical difference between both disaggregation emissions distributions. Coupling both vehicular emissions inventories into CMAQ and comparing the model outputs with observational data demonstrated that the proposed weighting factors model enhances the accuracy of PM_{10} simulations. While the weighting factors methodology provides a more accurate estimation for PM_{10} , it shows mixed results for pollutants like NO_2 and CO, where emissions are affected by finer-scale variations in traffic patterns and atmospheric chemistry. Additionally, employing different vehicular emissions inventories as input into CMAQ does not influence the ozone modelling results. This suggests that future studies could refine emission models by incorporating factors specific to pollutant characteristics, enhancing the model's ability to capture these variations.

The findings of this study have important implications for environmental planning and policy-making. The enhanced spatial resolution of vehicular emissions data provided by the weighting factors methodology can support urban planners in identifying high-emission zones more accurately, enabling targeted interventions in areas with high pollution exposure. For instance, this method could be used to identify major emission sources along highways and primary roads in densely populated regions, allowing for strategic placement of green infrastructure [1], low emission zones [22], or traffic restrictions [5] to reduce pollutant concentrations near residential areas. Moreover, the improved accuracy in PM_{10} predictions can aid policymakers in developing more effective air quality strategies, especially in metropolitan regions where particulate matter is a significant concern. By applying this methodology, environmental agencies can better prioritise actions to reduce PM_{10} in critical areas, potentially informing regulations that encourage cleaner vehicle technologies or stricter emissions standards for specific vehicle categories.

This study demonstrates the potential for improving the spatial resolution of vehicular emissions using widely available data, such as the proportion of primary road length by municipality. Future work could focus on incorporating high-resolution traffic datasets, such as hourly vehicle counts or GPS-based traffic flow data, to enhance the predictive accuracy of the GLM in different urban settings. Additional data sources, like satellite-derived emission estimates or air quality sensor networks, could be integrated to further refine weighting factors. Testing this method in diverse regions, such as rapidly urbanising cities, would help assess its adaptability across various traffic patterns and infrastructure layouts. Additionally, comparing the proposed weighting factors disaggregation method with other advanced disaggregation methodologies, such as land-use regression models or machine learning-based approaches, could offer insights into its relative performance.

Despite these promising results, a few key limitations should be acknowledged. First, the method demonstrated limited predictive accuracy for NO_2 and CO emissions, likely due to the complex and localised variations in traffic patterns and atmospheric chemistry that affect these pollutants. The model's reliance on primary road length as the independent variable may not fully capture the spatial nuances needed for pollutants that exhibit high variability near emission sources. Furthermore, our method was tested primarily in Brazilian metropolitan areas; therefore, its applicability to regions with different urban structures, traffic compositions, or environmental regulations remains uncertain. Future research could address these limitations by incorporating additional road-level and temporal data to enhance the model's robustness across diverse settings.

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